

View results

Respondent

29

Anonymous

101:29

Time to complete

A. General information about the respondent

1. * Country

Thailand



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Higher Education, Science, Research and Innovation

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

www.mhesi.go.th

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

Ministry of Higher Education, Science, Research and Innovation (MHESI) is a government agency responsible for Thailand's development of higher education, science, research and innovation. MHESI oversees 154 higher education institutes and 16 research and STI related organizations, as well as working across ministries to support research and innovation to other ministries' functions. MHESI supports higher education institutions's improvement of their academic capabilities and boost efficiency of R&D budget spending and promote the development of high technologies and innovation.

7. * Full name of officially designated contact person(s) who completed this survey

Nongnuch Chunbandhit

8. * Position of officially designated contact person(s) who completed this survey

Director, International Cooperation Strategy Group, International Affairs Division, MHESI

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Luksmon Smansin, Ph.D., Director, International Affairs Division, MHESI

11. Brief description of the consultation process established for the preparation of the report

Reporting and sharing of relevant and useful information in the scope of the organisation's work and responsibilities as open as possible and as appropriate, in relation to the implementation of the 2021 Recommendation on Open Science.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Office of National Higher Education, Science, Research and Innovation Policy Council

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

www.nxpo.or.th

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☒

Public

☐

Private

☐

Civil Society

☐

Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒

Yes

☐

No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Office of National Higher Education, Science, Research and Innovation Policy Council (NXPO),
MHESI

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Thai language, 2021 Recommendation on Open Science is available in concise versions disseminated in governmental journals and on public websites, e.g.
<https://www.nstda.or.th/rqm/rcr-news/177-rcr-news-07-open-science.html>,
http://www.bic.moe.go.th/images/stories/BIC_Mag_Apr-Jun_65.pdf.

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

There have been forums, seminars and open multi-stakeholder discussions on open science benefits and its real and apparent challenges to support the dissemination of scientific knowledge to scholars in other research fields, decision makers and the public at large, e.g. on 11 November 2024, the National Higher Education, Science, Research and Innovation Policy Council under the Ministry of Higher Education, Science, Research and Innovation, in collaboration with Thailand Science Research and Innovation (TSRI), Program Management Unit for Competitiveness (PMU-C), Program Management Unit for Human Resources & Institutional Development Research and Innovation (PMU-B) National Innovation Agency (NIA), National Science and Technology Development Agency (NSTDA), The Federation of Thai Industries (FTI), Mahidol University, and the Thai Academy of Science and Technology (TAST) hosted the International Synthetic Biology Conference in Thailand 2024 or SynBio Consortium 2024 in Bangkok. This event gathered leading experts, researchers, and innovators from around the world to discuss the possibilities of synthetic biology, share experiences, promote collaboration, and exchange knowledge to address global challenges such as healthcare innovation, sustainable agriculture and clean energy.
<https://www.nxpo.or.th/th/en/29333/>

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Since its inception, the Office of National Higher Education, Science, Research and Innovation Policy Council (NXPO) has supported the creation, co-production with multiple actors and the dissemination of responsible research and scientific knowledge to researchers, scholars and the public at large via NXPO website (<https://www.nxpo.or.th/th/en/>), media and networks. Additionally, NXPO continually invests in open science infrastructures, services and platforms, i.e. Talent Thailand: High-Potential Manpower Utilization Platform which is a Talent Utilization Platform focused on processing and disseminating research data for agencies involved in the production and development of manpower, such as schools, universities, research institutes, and government agencies, to be able to plan effective utilization of skilled workforce. For further information, please visit <https://talent.nxpo.or.th/>.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.



Yes



No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Higher Education, Science, Research and Innovation Policy: <https://www.nxpo.or.th/th/en/higher-education-science-research-and-innovation-policy/>; National Science, Research and Innovation Plan 2023 – 2027: <https://www.nxpc.or.th/content/documents/8181>

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☒ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Higher Education, Science, Research and Innovation Policy: <https://www.nxpo.or.th/th/en/higher-education-science-research-and-innovation-policy/>; and National Science, Research and Innovation Plan 2023 – 2027: <https://www.nxpc.or.th/content/documents/8181>

29. 2.3 In your country, are there specific policy instruments^[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☒ Yes
- ☐ Under development
- ☐ No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Policy Innovation Platform for the Better Future (PIP) is a central platform that opens for brainstorming and crowdsourcing. The aim is to bridge the gap between policymakers and people in the policymaking process to effectively tackle disruptive problems, e.g. the impact of COVID-19 and the effects of learning loss in higher education. It provides opportunities for the public to participate in policymaking process to obtain new policy innovations and creates a network for those that are ready to bring the policy into actions after their policies have been selected. Utilizing a prize or competition-based model to incentivize new policy ideas, PIP creates the window of opportunity for the public agencies, innovators and stakeholders to co-create policy solutions and overcomes traditional silos of policymaking process. Allocated budget for PIP is 1,000,000 THB/year. Thailand Higher Education and Innovation Policy Accelerator (THIPA) is an innovation policy platform aimed at accelerating the policy-making process in an organization in order to respond to changes and uncertainties in a timely manner. The platform is set up to assist other organizations to design appropriate policies that can drive innovation in government, academia, businesses, and society, using the best available tools for both analysis and action. THIPA works on inter-agency projects that involve higher education, science, research and innovation policy and have a scale-up potential to make a big impact on society. Proposals are subject to a readiness assessment prior to proceeding. Allocated budget for THIPA is 4,000,000 THB/year While PIP is a platform that identifies issues corresponding to the public's needs, THIPA turns the policy into action through engagement with stakeholders. In order to address the unmet needs, during a policy design process, THIPA brings together policymakers and stakeholders to co-develop new solutions. The design thinking tools and process are employed to address the challenges in policy implementation, and expected to reach a suitable and actionable policy for all parties involved. Website/Link to more information about the practice: <https://pip.nxpo.or.th/> and <https://www.nxpo.or.th/th/en/thailand-innovation-policy-accelerator/>

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)



Yes



Under development



No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Open science has been incorporated at the national level in the National Science, Research and Innovation Plan 2023 - 2027: <https://www.nxpc.or.th/content/documents/8181>.

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

In addition to the production of knowledge with multiple actors in various forms including policy papers and recommendations, white papers, publications, forums, seminars and discussions since its inception (<https://www.nxpo.or.th/th/en/report-category/printed-media-en/>), the Office of National Higher Education, Science, Research and Innovation Policy Council (NXPO) continually invests in open science infrastructures, services and platforms, e.g. Talent Thailand, High-Potential Manpower Utilization Platform, Policy Innovation Platform for the Better Future (PIP), Thailand Higher Education and Innovation Policy Accelerator (THIPA). For further information, please visit <https://talent.nxpo.or.th/>; <https://pip.nxpo.or.th/> and <https://www.nxpo.or.th/th/en/thailand-innovation-policy-accelerator/>

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.



Yes



Under development



No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

Program Management Unit on Area Based Development (PMU-A): <http://pmua.or.th> Program Management Unit for Human Resources & Institutional Development and Research and Innovation (PMU-B): <https://pmu-hr.or.th/en/home/> and Program Management Unit for Competitiveness (PMU-C): <https://pmuc.or.th/en/home/>

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

For example, PMU-C has integrated open science practices in existing research funding mechanism of which the mission is to administer research fund to enhance the nation's competitiveness by fostering collaboration between the public and private sectors in Thailand and from abroad. <https://pmuc.or.th/en/about-2/vision-2/>.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☐ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

40. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

Thailand's gross expenditures on research and development (GERD) in 2022 accounted for 1.16% of the country's GDP. <https://datacenter.nxpo.or.th/dashboard>

41. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

There were 63.21 million internet users in Thailand at the start of 2024, when internet penetration stood at 88.0 percent. <https://datareportal.com/reports/digital-2024-thailand>

42. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

43. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Thailand Research Education Network (ThaiREN) was established in 2005 to coordinate among research and education networks in Thailand as well as collaborate with international R&E networks. <https://www.thairen.net.th/>

44. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

45. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☐ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☒ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☒ community-based monitoring and information systems to complement national, regional and global data and information systems

46. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Talent Thailand is a Talent Utilization Platform focused on processing and disseminating research data for agencies involved in the production and development of manpower, such as schools, universities, research institutes, and government agencies, to be able to plan effective utilization of high-level manpower. Talent Thailand integrates work between related departments in designing, collecting and analyzing data to formulate human resource development policies. The platform is a data analysis system that includes information on high-potential workforce, both residing domestically and abroad, covering 10 target industries including cutting-edge research such as quantum technology, space technology, energy physics, social sciences, humanities, fine arts, music, sports and advanced arts. For more information: <https://talent.nxpo.or.th> Policy Innovation Platform for the Better Future (PIP) is a central platform that opens for brainstorming and crowdsourcing. The aim is to bridge the gap between policymakers and people in the policymaking process to effectively tackle disruptive problems. It provides opportunities for the public to participate in policymaking process to obtain new policy innovations and creates a network for those that are ready to bring the policy into actions after their policies have been selected. Utilizing a prize or competition-based model to incentivize new policy ideas, PIP creates the window of opportunity for the public agencies, innovators and stakeholders to co-create policy solutions and overcomes traditional silos of policymaking process. For more information: <https://pip.nxpo.or.th/> Thailand Higher Education and Innovation Policy Accelerator (THIPA) is an innovation policy platform aimed at accelerating the policy-making process in an organization in order to respond to changes and uncertainties in a timely manner. The platform is set up to assist other organizations to design appropriate policies that can drive innovation in government, academia, businesses, and society, using the best available tools for both analysis and action. THIPA works on inter-agency projects that involve higher education, science, research and innovation policy and have a scale-up potential to make a big impact on society. Proposals are subject to a readiness assessment prior to proceeding. For more information: <https://www.nxpo.or.th/th/en/thailand-innovation-policy-accelerator/>

47. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☐ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

48. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

49. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

To support high-skilled workforce development to meet industry demands, and enhance the national competitiveness, the STEMPlus Platform is designed to cater to the needs of enterprises for a skilled workforce as well as individuals interested in skill and competency development for career opportunities, enabling universities to develop educational programs that equip students with required skills and competencies. The STEMPlus Platform is established to bring together industry, academia and government to develop skilled workforces to meet the demands of the current and future labor market. Since its inception, the platform has certified over 800 training courses contributing to the upskilling of 80,000 workers, and 5,725 STEM employments, both represent key incentives under the Thailand Plus Package. The numbers of STEM employments and upskilled workers are expected to reach 10,000 and 100,000, respectively by the end of this year (2024). For more information: <https://www.nxpo.or.th/th/en/19266/>; <https://www.nxpo.or.th/th/en/23726/>; <https://www.nxpo.or.th/th/en/19255/>; <https://www.nxpo.or.th/th/en/28140/>; and <https://www.stemplus.or.th/>

50. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☒ Yes

☐ No

51. 4.2 cont. **If yes**, please provide more information and links as relevant.

Policy Innovation Platform for the Better Future (PIP) is a central platform that opens for brainstorming and crowdsourcing. The aim is to bridge the gap between policymakers and people in the policymaking process to effectively tackle disruptive problems. For more information: <https://pip.nxpo.or.th/>. Thailand Higher Education and Innovation Policy Accelerator (THIPA) is an innovation policy platform aimed at accelerating the policy-making process in an organization in order to respond to changes and uncertainties in a timely manner. The platform is set up to assist other organizations to design appropriate policies that can drive innovation in government, academia, businesses, and society, using the best available tools for both analysis and action. THIPA works on inter-agency projects that involve higher education, science, research and innovation policy and have a scale-up potential to make a big impact on society. For more information: <https://www.nxpo.or.th/th/en/thailand-innovation-policy-accelerator/>.

52. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☒ Yes

☐ No

53. 4.3 cont. **If yes**, please provide more information and links as relevant.

The Higher Education Plan for Manpower Planning and Development 2023-2027 and the Higher Education Plan 2021-2027: <https://www.ops.go.th/th/aboutus/strategic-policy/item/8417-2566-2570>

54. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☐ Partly
- ☐ No

55. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

56. 4.5 cont. **If yes**, please provide more information and links as relevant.

FameLab Thailand was launched in 2015 with the great and continuous support from Ministry of Science and Technology (currently Minister of Higher Education, Science, Research and Innovation), universities, and media partners. The programme aligned with the national priorities on strengthening capacity of academics and promoting importance and awareness of science. The results and outcomes were vast; from education opportunities and career development, research collaborations, initiating science communications courses in universities, spin-off to School-equivalent competition - School Lab Thailand, to the FameLab Thailand project being awarded with the Mahidol Award for Science Communications 2021. For more information: <https://www.britishcouncil.or.th/en/famelab/about-competition>.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

57. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (*ref.: (v) 20.b, 20c, 20i*)

☐ Yes

☐ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

58. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☒ YES, at both the national and institutional levels
- ☐ No

59. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☐ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

60. 6.1. cont. **If yes**, please provide more information and links as relevant

E-Participation for Citizen-Centric Policymaking” is a virtual platform designed to engage the public in the policy design process. The platform was built upon “Policy Innovation Platform for the Better Future” which was established by the Office of National Higher Education, Science, Research and Innovation Policy Council (NXPO) for crowdsourcing policy innovation during the COVID-19 pandemic. For more information: <https://www.nxpo.or.th/th/en/23461/>

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

61. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Organize multistakeholder forums, conference and discussion; join the global research centers, organizations, forums, consortiums; collaborate with the international organizations including UN organs and UNESCO and global partners to create knowledge and bridge the digital, technological and knowledge gaps, etc.

62. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☐ No

Part 8. GENERAL CONSIDERATIONS

63. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

64. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org